Exercise 02 SQL Foundations

SQL Aggregate functions & operators

1. Distinct departments

```
SELECT DISTICT department
FROM students;
```

department
IT
HR
Finance

2. Average age per department

```
SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

3. Department with more than 1 student

```
SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY students
HAVING COUNT(*) > 1;
```

department	student_count
IT	2
HR	2

4. Students whose age is between 21 and 23

```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. IT or HR department who are older than 21

```
SELECT student_id, name, age, department
FROM students
WHERE department IN('IT','HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	EVE	22	HR

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6. Credits per department, only for departments with more than 5 total credits

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

7. Courses that do not have 4 credits

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;
```

course_id	course_name	department	credits
101	SQL	IT	3
104	Excel	Finance	2
105	statistics	HR	3

8. Top 3 courses by credits in descending order

```
SELECT courses_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. max, min, and avg grade across all enrollment

```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade,
~~FROM~~ FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84,6

10. Count how many enrollments exist per course

```
SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

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11. total salary and total bonus per department

```
SELECT department,
       SUM(salary) AS total_salary;
       SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10500
HR	109000	7500
Finance	70 000	6000

12. departments where avg salary is above 55000

```
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61 000
Finance	70 000

13. employees whose salary plus bonus is greater than 60 000

```
SELECT employee_id, name, salary, bonus,
       (salary + bonus) AS total_compensation
FROM salaries
Where (salary+bonus) > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65 000
3	Spike	70000	6000	76 000
4	Tyke	62000	5500	67 500

14 Only include departments with average budget above 70 000

```
SELECT department,
       SUM(budget) AS total_budget
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000
```

department	totalbudget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. budget between 50,000 and 120000, excluding the marketing department

```
SELECT project_id, Project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120 000
AND department <> 'marketing';
```

Project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000

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